

AI · WEATHER · HEALTH · AQI

AEROHEALTH AI

AI-Powered Personalized Weather & AQI Health Advisory

"The environment is the same. The risk is not."



Temperature

34°C

Feels Like 37°C



AQI Index

165

Unhealthy

TEAM NAME

HACK TITANS

**COLLEGE /
INSTITUTION**

VIT BHOPAL



Personal Risk

72 / 100

HIGH

**HACKATHON /
EVENT**

ORIGIN

**PROBLEM
STATEMENT**

PS-4

 Environmental Intelligence ·  AI Advisory ·  Personalized

Made with **GAMMA**

THE PROBLEM

Why Generic Environmental Alerts Fail

Imagine two people standing in the **same city**, at the **same moment**, breathing the **same air**. They both receive the exact same alert.

 PERSON A – Healthy Ad Indoor Worker · No Conditions • Works in an air-conditioned office • Minimal outdoor exposure • No respiratory sensitivity		
 Temp 34°C	 AQI 165	 PM2.5 82 µg/m³
 PERSON B – Asthma Patient Outdoor Worker · Respiratory Condition • Works outdoors all day • Prolonged PM2.5 exposure • Asthma increases sensitivity		
 Both people receive: "Air quality is poor. Limit outdoor exposure."		
Person A's Reality Lower actual exposure indoors. Generic alert may be overly cautious.	Person B's Reality Respiratory sensitivity + prolonged outdoor exposure = significantly higher personal risk. Alert is dangerously under-specific.	The Core Gap No alert system accounts for who you are when telling you about the air around you.

EXAMPLE USER · ILLUSTRATIVE DATA

Meet Rahul – Our Example User

 All values below are **illustrative prototype data** used to demonstrate system flow. Not real-world measurements.

USER PROFILE

Name

Rahul

Age Group

Adult

Health Condition

Asthma 

Occupation

Outdoor Worker 

Location

Ashta, Madhya Pradesh 

 RISK OUTPUT

72 / 100
HIGH RISK

↑ Profile + Environment
→ Personalized Risk

LIVE ENVIRONMENTAL DATA

 Temperature

34°C · Feels Like 37°C

 Humidity

68%

 Wind

2.4 m/s

 AQI

165 – Unhealthy

 PM2.5

82 µg/m³

 PM10

138 µg/m³

Data Enters the System → Risk Is Calculated

ENVIRONMENTAL INTELLIGENCE ENGINE – INPUT



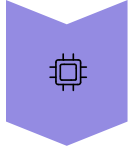
PERSONAL INPUTS

Age: Adult · Health: **Asthma** · Occupation: **Outdoor Worker** · Location: Ashta



ENVIRONMENTAL INPUTS

Temperature: **34°C** · AQI: **165** · PM2.5: **82** · PM10: **138**

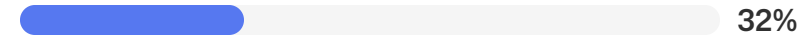


PERSONALIZATION ENGINE

Combines Environmental Severity + Personal Sensitivity + Exposure Context

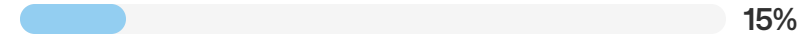
ILLUSTRATIVE PROTOTYPE SCORING – Rahul's Risk

☐ This is a prototype scoring example, not a medically validated formula.



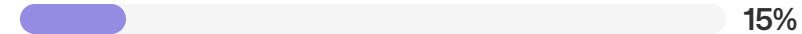
AQI Factor

AQI 165 – Elevated pollution level



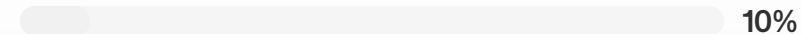
Temperature Factor

34°C heat stress contribution



Health Sensitivity

Asthma increases vulnerability



Occupation Exposure

Outdoor work = prolonged exposure

⚠ TOTAL PERSONAL RISK SCORE

72 / 100 – HIGH RISK

AQI Factor (32) + Temperature (15) + Health Sensitivity (15) + Occupation (10) + Age (0) = 72 / 100

This Is Where Personalization Matters

Same weather. Same AQI. Same city. **Three completely different risk levels and recommendations.**

 **Temperature: 34°C** ·  **AQI: 165** ·  **PM2.5: 82 µg/m³** – Applied to all three users identically

 **USER 1**
Healthy Adult
Indoor Worker
No Health Conditions

Illustrative Risk

42

/ 100 · MODERATE

Indoor work + no sensitivity = lower exposure factor

 **USER 2**
Asthma Patient
Indoor Worker
Respiratory Condition

Illustrative Risk

57

/ 100 · ELEVATED

Asthma sensitivity adds risk despite indoor work

 **USER 3 – Rahul**
Asthma Patient
Outdoor Worker
Respiratory + Exposure

Illustrative Risk

72

/ 100 · HIGH

Asthma + outdoor exposure = highest personal risk

Same Weather + Same AQI Environmental conditions are identical for all three	Different Health + Different Exposure Profile drives the personalization engine
Different Risk Score 42 → 57 → 72 based on profile	Different Recommendation Each person receives advice calibrated to their actual situation

This is the core innovation of EnviroGuard AI.

What Does AI Actually Do?

The AI is **not** used merely to display AQI numbers. It converts structured environmental + personal data into **simple, personalized, action-oriented language**.

STRUCTURED CONTEXT BUILT BY THE SYSTEM

User

Rahul

Profile

Asthma + Outdoor Worker

Environment

34°C + AQI 165 + PM2.5 82

Risk Score

72 / 100 – HIGH

↓ Passed to LLM



Groq / Gemini Free-Tier LLM

Structured input → Natural language advisory output

INPUT TO AI

"Adult + Asthma + Outdoor Worker

+ 34°C + AQI 165 + PM2.5 82

+ Risk Score: 72/100 (HIGH)"

↓ AI PROCESSING ↓



PERSONALIZED OUTPUT

"Current air quality is elevated and the temperature is high. Because you have asthma and work outdoors, prolonged exposure may increase your environmental risk. Consider reducing strenuous outdoor activity when possible, taking regular breaks in cleaner indoor areas, staying hydrated, and checking conditions again before extended outdoor work."

Key Capability

AI converts structured environmental + personal data into **understandable, action-oriented advice** – not just raw numbers.

Under the Hood



USER PROFILE INPUT

Age · Health Condition · Occupation · Location



WEATHER DATA – Open-Meteo API

Temperature · Humidity · Wind Speed · 7-Day Forecast



AIR QUALITY – WAQI / Govt AQI Dataset

AQI · PM2.5 · PM10 · Pollutant Breakdown



DATA PROCESSING LAYER

Parse API responses · Normalize values · Combine environmental inputs



PERSONALIZATION ENGINE

AQI Factor + Temperature + Health Sensitivity + Occupation + Age → Risk Score



AI LAYER – Groq / Gemini LLM

Structured context → Personalized advisory text



STREAMLIT DASHBOARD OUTPUT

AQI · Weather · Risk Meter · AI Advisory · 7-Day Trends



SYSTEM STATUS INDICATORS



WEATHER API

CONNECTED · Open-Meteo · Real-time temperature, humidity, wind, forecast



AQI API

CONNECTED · WAQI / Govt Dataset · AQI, PM2.5, PM10



RISK ENGINE

ACTIVE · Personalization scoring · Multi-factor calculation



AI ENGINE

READY · Groq / Gemini LLM · Advisory generation



TECH STACK

Frontend

Streamlit (Python)

APIs

Open-Meteo · WAQI

AI

Groq / Gemini Free Tier

Engine

Python · Custom Risk Logic

What Rahul Sees on the Dashboard

ⓘ ⚠ All values shown are **illustrative UI prototype data** to demonstrate dashboard layout and functionality.

🌤 WEATHER PANEL

Temperature
34°C · Feels Like 37°C

Humidity
68%

Wind Speed
2.4 m/s

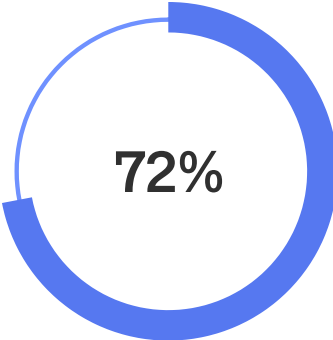
🌬 AIR QUALITY PANEL

165
AQI – UNHEALTHY

PM2.5
82 µg/m³

PM10
138 µg/m³

⚠ PERSONAL RISK METER



72%

Personal Risk Score
HIGH RISK
Adult · Asthma · Outdoor Worker

👤 USER PROFILE

Adult · **Asthma** · Outdoor Worker
Ashta, Madhya Pradesh

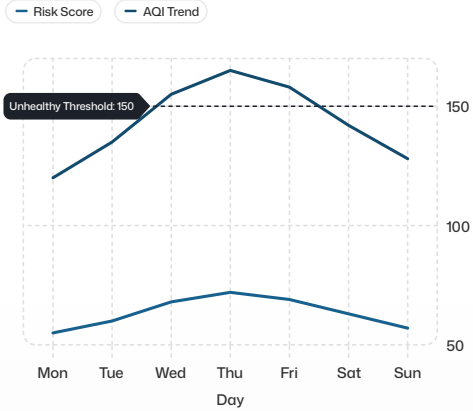
🌅 BEST OUTDOOR TIME

Recommended Window
Early morning before 7:00 AM or after sunset when AQI typically improves

🤖 AI ADVISORY

"Current air quality is elevated and the temperature is high. Because you have asthma and work outdoors, prolonged exposure may increase your environmental risk. Consider reducing strenuous outdoor activity, taking regular breaks in cleaner indoor areas, staying hydrated, and checking conditions before extended outdoor work."

📈 7-DAY TREND



Illustrative trend data only

EnviroGuard AI vs. Traditional Apps – & Who Benefits

FEATURE COMPARISON

Traditional Weather / AQI App	EnviroGuard AI
Shows temperature	Shows temperature 
Shows AQI	Shows AQI 
Generic warning for all	Personalized advisory per user 
Same alert for everyone	User-specific recommendation 
No personal context	Age + Health + Occupation 
Data-focused output	Decision-focused output 
"AQI is 165"	"Here is what AQI 165 means for YOU" 

**We are not replacing environmental data.
We are adding personal intelligence to it.**

WHO CAN BENEFIT?

**Asthma / Allergy Users**

AQI elevated → Higher sensitivity → More cautious, specific recommendation

**Outdoor Workers**

Long exposure factor → Frequent breaks + exposure reduction guidance

**Students / Athletes**

High temp + poor AQI → Outdoor activity timing recommendation

**Elderly Users**

Environmental stress → More conservative, protective advisory

 **The Core Logic**

Personal Profile → Personal Risk → Personal Action

From Environmental Dashboard to Personal Environmental Intelligence

PRODUCT ROADMAP



FUTURE CAPABILITIES



Mobile Notifications

Smart alerts based on your profile + forecast



Hyperlocal AQI

Street-level pollution intelligence



Wearable Integration

Real-time exposure tracking via smartwatch



Voice AI

Speak to your environmental health advisor

*"Don't just measure the environment.
Personalize what it means."*

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